

# Larry Nguyen

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## Education

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University of Maryland, College Park, MD

Anticipated Graduation: Dec. 2027

**B.S. of Electrical Engineering**

**Relevant Coursework:** Advanced Analog and Digital Electronics, Digital Logic Design, Electric Circuits, Electromagnetics, Introduction to Micro and Nanoelectronics, Signal & System Theory

## Experience

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**Pho An & Grill - Silver Spring, MD**

Jun. 2025 – Present

*Server & Busser*

- Served customers, handled food service, and helped maintain smooth operations during busy shifts
- Trained new employees on restaurant procedures and day-to-day service

**Navy Yard - Washington, D.C.**

Sept. 2023 – May 2026

*Shadow | Boiler and Chiller Operations*

- Shadowed boiler and chiller operators; assisted with replacing faulty contactors, pulling wire, and diagnosing components showing 0 V
- Assisted with troubleshooting basic electrical and low-voltage equipment under operator supervision.

**UMBC Mechanical Rooms - Baltimore, MD**

Jun 2025 – Oct. 2025

*Shadow | Boiler and Chiller Operations*

- Observed VFDs, relays, and contactors in mechanical cooling/heating equipment, using multimeters and non-contact voltage testers.

## Projects

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**Adjustable Linear Bench Power Supply** | *Personal Project - In Progress*

- Designed a 0–15V, 1.2A linear regulated power supply, including rectification, filtering, relay protection, and voltage regulation stages.
- Simulated in LTspice before layout; completed PCB layout and fabrication, now preparing for hardware testing
- Diagnosed and resolved a hard fault on regulator output by tracing the failure through the regulation chain.

**DC-DC Buck Converter with PID Control** | *Personal Collaborative Project*

- Performed calculations, ran simulation, and PCB layout for a 12V-to-5V, 2A synchronous buck converter, plus a secondary voltage divider to step output down to 3.3V for MCU compatibility
- Selected gate drivers, MOSFETs, and RLC components for 101 kHz switching operation
- Identified a PCB footprint mismatch for the feedback resistor causing a 3.21V output against the 3.3V target; corrected with an off-board resistor mount to validate the fix ahead of the next revision

**ENES 100 Project** | *Semester Project*

- Worked in a team of six to design and build an autonomous vehicle that identified a target, measured its dimensions, and navigated a course.
- Led electronics for sensors, actuators, and microcontroller control; awarded Most Creative OTV among six teams.

**Arcade Machine Restoration & Modernization Project** | *Personal Project*

- Independently upgraded a vintage arcade cabinet to support over 60,000 retro games by integrating a Raspberry Pi 5 running Batocera Linux.
- Removed obsolete arcade hardware and rewired 15 LED-illuminated buttons and joysticks to a USB encoder board, integrating the control panel with a Raspberry Pi 5

## Skills

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**Hardware:** Arduino, KiCad, LTspice, Circuit Design, Soldering, Schematics, Electronics, Basic Electrical Equipment

**Software:** MATLAB, Verilog, MIPS Assembly, GitHub, Excel, Autodesk CAD